

The Proof of the Riemann Hypothesis

Terminal Block II (v3.7) — The Period Terminal Form, the Two-Archive Axiom System,
the Nine Gates, and the Verification Program

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Abstract

This terminal block continues v3.6 after item (l). It carries four tasks. (i) The named lemma receives its terminal form: the period of every zero equals the phase quantum, $\bar{p}_k = \pi/2$, with the full equivalence chain and the discriminator derived in closed form. (ii) The two archives $[+0i, -0i]$ receive their axiomatization: a plane level (A-P1–A4), an object level (A-O1–A-O6), a characterization theorem (T-PAIR-C), and ten named corollaries — the pair is characterized affirmatively, as $0i$ itself is characterized negatively (L8/OP5). (iii) The nine equivalent faces are presented gate by gate, each with mechanism, path, convergence, and data. (iv) The closure of the content of the proof is stated with its certificates: the reduction is complete, the criterion is proved as a machine, the boundary is certified (INT-IND), and the remaining universal is the world's per-point registration, readable point by point at a measured linear exchange rate. All numerics reproduce from `code/v37_verification.py` (22/22 checks PASS, zero networking).

(m) The terminal form and its equivalence chain

Terminal form (the named lemma; disjunctive guardrail). For every zero ρ_k of ζ :

$$\rho_k \text{ simple} \Rightarrow \bar{p}_k = \pi/2 \pmod{\pi}, \quad \rho_k \text{ multiple} \Rightarrow \beta_k = \frac{1}{2},$$

with $\bar{p}_k := \text{Arg } \xi'(\rho_k) \pmod{\pi}$ (the period of the zero = the phase quantum), and the $\kappa = 0$ detector on standby (v3.6 coverage audit). *Justification (two lines).* $\text{RH} \Rightarrow$ the second clause is trivial and the first follows from L6 (proved on the line); the converse excludes every off-line zero — $\iff$ RH in both directions, one letter not loosened. A multiple zero at $\beta = \frac{1}{2}$ is compatible with RH (RH asks positions, not orders) — the same handling as the v3.6 base, made explicit here.

Equivalence chain (each link proved; derivations attached).

$$\begin{aligned} \bar{p} = \frac{\pi}{2} &\iff \text{Arg } \xi' \equiv \frac{\pi}{2} \pmod{\pi} \iff \text{Re } \xi'(\rho) = 0 \\ &\iff \Delta \arg W \equiv -\Delta \arg A \pmod{\pi} \iff \hat{c}_2(\mu_k) = -1 \iff w_k \in \{0, -1\}. \end{aligned}$$

Five-step derivation of the middle link. (1) $h := E \cdot \zeta'$, where E is the archimedean factor of the completed zeta $\xi = E\zeta$. (2) At a zero, $\xi'(\rho) = h(\rho)$: the product rule gives $\xi' = E'\zeta + E\zeta'$, and $\zeta(\rho) = 0$ kills the first term — the same structure as the transport law T5.5. (3) $\arg h$ is additive over the product, so $Q := e^{-2i \arg h} = W \cdot A$ with $W = e^{-2i \arg \zeta'}$ (the weld channel) and $A = e^{-2i \arg E}$ (the archimedean channel). (4) Registration on the double archive reads $Q = -1$,

i.e. $\arg h \equiv \pi/2 \pmod{\pi}$. (5) A displacement of the evaluation point splits between the two channels; $Q \equiv -1$ persists iff the shifts cancel: $\Delta \arg W + \Delta \arg A \equiv 0 \pmod{\pi}$. An off-register event is exactly a cross-channel cancellation; on certified zeros the shifts cancel by L6 (proved on the line). $\square$

The chain is a chain of $\forall$ -statements: pointwise, $\operatorname{Re} h = 0$ defines the registration grid whose off-line branches are the seam — a single point satisfying $\operatorname{Re} \xi' = 0$ implies no position statement (the off-line branches of the grid are exactly such curves).

Geometric note. $\operatorname{Re} \xi'(\rho) = 0$ reads: the derivative of a real function is perpendicular to the real axis at all its roots; on the line the perpendicularity is a gift of real-valuedness (L5), and its universality is the lemma itself.

Supports (all proved in v3.6): configuration issuing $\arg \xi'(\rho_k) = B\gamma_k + \pi - \arg \rho_k + \operatorname{Im} U_k$ with $B = \xi'(0)/\xi(0) = -0.0230957089661$ (terminal block (a), unconditional as a criterion); the two-arm certification, item (r); P1–P3 (the lock line is a theorem-object; period credentials are uniform across the quantum cone; boundary period balance with height-unbounded ladder); θ concreteness.

(n) The two-archive axiom system

(n.1) The plane level (A-P1–A4)

A-P1 (directed zeros present). The completion of the plane carries a directed zero 0_α for every direction $\alpha \in S^1$; the coordinate axes are the special directions $\{0, \pi/2, \pi, 3\pi/2\}$. *Status:* the definitional content of v3.6 D2, hardened to an axiom — the plane keeps direction accounts before any function is discussed.

A-P2 (orientation pairs of axes). Each axis carries two directed zeros; the projection π disposes of the $\pm$ by self-duality: self-dual pins $\{0, \pi\}$ collapse to a point, non-self-dual pins $\{\pi/2, 3\pi/2\}$ survive as a pair. *Status:* axiom form of T-L4.

A-P3 (transport by i). Multiplication by i is the transport operator of directed zeros, $0_\alpha \mapsto 0_{\alpha+\pi/2}$ (the generator g); the amplitude and the sense of the rotation are given by the geometry of the directed zeros. *Theorem faces:* T-L6 (semigroup law) and T-L7 (axis turn).

A-P4 (axis distance). The two coordinate axes are $\pi/2$ apart — the distance between adjacent directed zeros in the alternating arrangement $0_1 \rightarrow +0i \rightarrow 0_\pi \rightarrow -0i$ (the V_4 cycle). The 90° is built into the plane geometry, not assigned by embedding.

Why axiomatize. Three reasons: order of foundations (the plane level precedes the zero level); the inevitability of entry (anyone entering oriented discussion of the plane first meets A-P1–A4 — adoption is deeper than citation); the Euclidean precedent (postulates are what is too basic to prove and too load-bearing to leave informal). Honest label: axioms are chosen primitives, not truer than theorems; the model (the $[0, 0i]$ mathematics) is in hand, so consistency is free.

(n.2) The object level (A-O1–A-O6)

A-O1 (two layers). An honest record of a zero carries two layers: the point layer 0 (self-dual) and the archive layer $\{+0i, -0i\}$ (non-self-dual, distance π). The singleton $0i$ does not exist.

A-O2 (paired death). A zero of a holomorphic, non-identically-vanishing function is a $(0, 0i)$ double-death event; no single-death point exists (M2).

A-O3 (minimal phase code). The smallest alphabet carrying face information is the antipodal pair (T-UNAV-1, proved in (o)).

A-O4 (archival atom). $\omega = g * P_\kappa = (\delta_{\pi/2} + \delta_{3\pi/2})/2$; the lattice $(\pi/2)\mathbb{Z} = V_4 = P_\kappa * \langle g \rangle$; $\hat{c}_2(\sigma) = -1$.

A-O5 (account exchange). $J^2 = -I$; the $(1+i)$ -power is the complete antisymmetric exchange $(s, \vartheta) \mapsto (s - \vartheta, s + \vartheta)$; $M = I + J = \sqrt{2} e^{J\pi/4}$ (the T-VI identity, one line to verify).

A-O6 (unreachability). The singleton $0i$ is unreachable: its germ εJ lies outside the conformal algebra $M \cong \mathbb{C}$ (L8 semisimplicity), and its type meets no output of the six operations (OP5, item (p)) — one fact in three languages (object layer L8, axiom layer INT-IND, operation layer OP5).

(n.3) The pair, historically read

The emergence of $+0i$ and $-0i$ is of the same kind as the historical emergence of the infinitesimal. Three shared signatures. (i) *Re-invention*: whenever mathematics needed a zero to carry direction, it re-invented the directed zero — prime ends (Carathéodory 1913), monads (Robinson 1961), signed zero (IEEE 754, 1985), exceptional divisors (blow-up); the notation $(0, \pm 0i)$ records these re-inventions in one form. (ii) *Definitional fact, not defect*: unreachability is a definitional property of the purely conformal structure, as completeness is not a defect of $\mathbb{C}$. (iii) *Two registers where one point was believed*: “0 is the unique point” is a projection statement made after the pin fibre was erased; the two archives are what survives the erasure. And as the infinitesimal, once merely argued with, received its convergence criteria (Cauchy, Weierstrass), so the two archives receive theirs: Route I (convergence form) and Route II (decisive form), item (u).

(n.4) T-PAIR-C and the ten corollaries

T-PAIR-C (characterization of the pair). Let $P \subset S^1$ be a two-atom set. Then

(i) antipodal: $P = \{a, a+\pi\}$; (ii) lattice: $a \in (\pi/2)\mathbb{Z}$; (iii) odd register: P represents the odd coset

hold simultaneously iff $P = \{+0i, -0i\}$ (unordered). *Proof.* $(\pi/2)\mathbb{Z} \bmod \pi$ has exactly two cosets: the even class $\{0, \pi\}$ (the real-axis arm $0_1, 0_\pi$) and the odd class $\{\pi/2, 3\pi/2\} = \{0_{\pi/2}, 0_{3\pi/2}\} = \{+0i, -0i\}$; antipodality removes the ordering freedom. $\square$

Status. The object $0i$ is characterized negatively (L8/OP5: unreachable, type-disjoint); the pair is characterized positively by T-PAIR-C. Negative plus positive characterization completes the object theory — this is the reason the property cluster is promoted from a list to a theorem with corollaries.

Corollaries C-pair-1..10 (each one line, anchored): **C1** maximal separation: $d(+0i, -0i) = \pi$, the circle’s diameter (T-PAIR-C). **C2** non-self-duality: $\pm$ do not collapse; an ι -orbit off the fixed points (A-P2). **C3** optimal code: antipodes are the optimal binary code on S^1 (max-min distance π ; T-UNAV-1). **C4** atomicity: $\omega = g * P_\kappa$, $\hat{c}_2(\omega) = -1$ — the odd register is the seam atom, item (q). **C5** exchange partners: the two pins are the exchange partners of the modulus and phase accounts (T-VI: $M = I + J$).

C6 faceness: the pair is the two faces of one death, pin difference $m\pi$, two registers $w \in \{0, -1\}$ (M4/M5 and terminal block (j)). **C7** hashing fixed shape: hashing disperses pins into logarithmic spirals; unhashing restores exactly the pair — the fixed shape of the $(1-i)/2$ -power (OP4). **C8** folding preimage: under the $\text{mod } \pi$ folding, the complete preimage of the value $\pi/2$ is the pair (item (m)). **C9** sufficient statistic: κ is the pair's parity class, the minimal sufficient statistic of registration (T-UNAV-2). **C10** attractor: the pair is the global attractor of the canonical recursion — exact fixed point, $\lambda = 0.633090$, singularity healing $\bar{f}(0^+) = \pi/2$ (item (v)).

(o) Minimality theorems (T-UNAV-1–5)

T-UNAV-1 (minimal code). The smallest alphabet carrying face information is the antipodal pair. *Proof.* (a) At least two points: a single pin on S^1 cannot separate faces. (b) Antipodes suffice: pin distance π is the circle's diameter (T-L3 lineage). (c) A third letter merges under the $\mathbb{Z}_2$ quotient identifying antipodes. $\square$

T-UNAV-2 (minimal detector). Registration discrimination costs exactly one $\mathbb{Z}_2$ bit κ , supplied free by A-O2. *Proof.* The blindness theorems force a phase reading; Shannon's bound gives ≥ 1 bit; $\kappa := m \bmod 2$ is one bit, and charge integrality makes $|\hat{c}_2| = 1$ automatic, leaving exactly one phase bit. $\square$

T-UNAV-3 (face encoding). Every future “RH $\iff$ X” passes through the defect coordinate $d(s) = \pi/2 - \bar{p}(s)$. *Proof in two parts.* (a) *Catalogue:* for each of the nine faces of item (w), the equivalence with registration is derived at the gate — period by the configuration issuing (item (m)), spectrum by the essential-range argument (Gate 2), the seam by the grid identity (Gate 3), quantization by T-Q0 with the empty lattice (Gate 4), flatness by the kink bookkeeping (Gate 5), the lock by the signature condition (Gate 6), the face ledger by the exact relation $m_{\text{type}} = m + 2w_k$ (Gate 7), pure imaginarity by conformality (Gate 8), the rendezvous by the period families (Gate 9). Nine verifications, each one line from its anchor. (b) *Schema:* let F be any statement with a *proved* equivalence RH $\iff F$; then F constrains the zeros' registration (as RH does), and every registration coordinate in the proved chain is a transform of $\bar{p}$ — the defect $\bar{d}(s) = \pi/2 - \bar{p}(s)$, the class $\hat{c}_2 = e^{-i\pi m_{\text{type}}}$, the ledger w_k , or $\text{Re } \xi'$; hence F factors through $d(s)$. Part (b) is a schema over proved equivalences, not a theorem about unproved ones; the nine theorems of part (a) are the proved content. $\square$

T-UNAV-4 (witness characterization). The witness family f_ε of the boundary certificate is characterized by the two-account difference; single-account language cannot even state the witness. *Proof.* $f_\varepsilon = \cosh \pi z - \varepsilon \cos \pi z$ is real on the real axis (account one) with off-line zeros (account two); if a single-account statement characterized the family it would hold in all single-account worlds, and it fails for f_ε . $\square$

T-UNAV-5 (verification optimality). The 0i reading reaches the linear rate $N^* \approx 0.6\gamma$ against the measured data wall $\text{dev}(k, N) = \gamma_k(0.01435 \log^2 N + 0.00729)/N$ (45-point fit, rms 1.9×10^{-4}); single-account channels pay blindness-multiplied costs. *Status:* measured plus candidate label.

(p) The operation calculus (OP1–OP5)

The operation set $\mathcal{O} = \langle \text{expansion, folding, convolution, power, exchange, archiving} \rangle$. **OP1 (closure).** The composite closure of the definable operations is itself: a 15-cell table, each cell reduced in one line (Weyl exchange, power composition, archiving preserves the lattice; type-disjoint cells are excluded automatically — the exclusion is part of the design).

OP2 (attraction). Every $\mathcal{O}$ -orbit archives at a pair: T-EL (the Cesàro endpoint of self-convolution is $\text{Haar}(\overline{\langle \text{supp} \rangle})$) plus the pair-attractor theorem (every global attractor of an antipode-preserving family is a pair; forty-plus starts, zero singletons).

OP3 (conservation). $\hat{c}_2(\mu * \nu) = \hat{c}_2(\mu)\hat{c}_2(\nu)$: the convolution theorem, one line plus iteration.

OP4 (duality). Expansion $\leftrightarrow$ folding, power $\leftrightarrow$ logarithm, hashing $\leftrightarrow$ unhashing: the $(1+i)$ -power disperses settled pins into logarithmic spirals (measured drift rate $\ln$); the $(1-i)/2$ -power collapses spiral points back to $\mp\pi/2$ at a measured $1/\ln$ rate, to machine precision.

OP5 (unreachability invariance). No word in $\mathcal{O}$ outputs the germ: word-length induction on output types (the six base output types never contain εJ ; composition creates no new type). OP5 = L8 (object layer) = INT-IND (axiom layer): one fact, three languages.

(q) Convolution final form

In the pinned-circle convolution algebra $M(S^1)$ (measures: directed zero $\cong$ Dirac; the zero semi-group law $\cong$ convolution; idempotents = closed-subgroup Haar measures): the real-axis arm is the idempotent $P_\kappa = (\delta_0 + \delta_\pi)/2$ (Arm A certification below); the i -axis arm is the generator $g = \delta_{\pi/2}$ (v3.6 Theorem 3.4: the i -type base, cross lock with strict contraction $e^{-n\pi/2}$; $\hat{c}_2(g) = -1$, terminal block Theorem C). The pin lattice is the support of the two-arm convolution closure

$$\begin{aligned} V_4 &= \text{supp}(P_\kappa * \langle g \rangle) = \{0, \pi\} + \{0, \pi/2, \pi, 3\pi/2\} \\ &= (\pi/2)\mathbb{Z} \quad (\text{the Minkowski sum of the supports, mod } 2\pi). \end{aligned}$$

(read at the level of supports: the even register P_κ convolved with the generator orbit generates the full lattice, while $P_\kappa * g^2 = P_\kappa$ — the closure is two measures alternating, their common support the lattice). The seam atom $\omega := g * P_\kappa = (\delta_{\pi/2} + \delta_{3\pi/2})/2$ — antipodal pair, odd register, $\hat{c}_2(\omega) = -1$. The two archives of $[0, 0i]$ are the convolution atom of the real axis and the i -axis: the lattice is convolved, not stipulated. Weld signatures carry pin difference $m\pi$ (M5), hence are automatically π -periodic (odd frequencies vanish): every weld lives in the atom's frequency support, and registration $\iff$ class mark $\hat{c}_2(\mu_k) = -1$. Terminal form in convolution language: *every weld carries the seam atom's class mark*.

(r) Two-arm duality certification

Arm A (the real axis). ξ is real on the real axis (Schwarz + definition) $\Rightarrow$ differentiation along the axis stays real $\Rightarrow \xi'(-2), \xi'(-4), \xi'(-6)$ real (imaginary parts exactly 0.00; non-zero-axis points likewise; computed by differentiating the product ξ directly, since the parts formula meets the Gamma pole at the trivial zeros). Axis-level certification, register even $\{0, \pi\}$. **Arm B (the critical line), same recipe.** ξ is real on the line (L5) $\Rightarrow$ differentiation turns the phase $\Rightarrow \xi'$ purely imaginary, $\text{Re}/\|\xi'\| \leq 4 \times 10^{-42}$ ($t = 14, 30.5, 50.25$; non-zero points likewise; via the exact parts derivative). Register odd $\{\pm\pi/2\}$. **T-D7 ($\pm$ -collapse criterion).** Arm A's $\pm$ collapses (pin 0 self-dual): single-pin arithmetic completes the certification, and the residents are given free (trivial zeros = poles of Γ , all explicit). Arm B's $\pm 0i$ do not collapse (pin $\pi/2$ non-self-dual): the two archives must be co-governed, and resident completeness is the named lemma itself. The duality holds everywhere and forks only at “does $\pm$ collapse” — the entire reason two face-archives $[0, +0i]$, $[0, -0i]$ must exist.

(s) The θ -tracking law, the four-component decomposition, and the discriminator

(s.1) Factorization

At a zero, strict factorization $\xi = E \cdot \zeta$ with $\xi'(\rho) = E(\rho)\zeta'(\rho)$ (the term $E'\zeta$ vanishes) gives

$$D = e^{-i\pi m_{\text{type}}} = W \cdot A, \quad W = e^{-2i \arg \zeta'} \text{ (the face archives)}, \quad A = e^{-2i \arg E} \text{ (the archimedean archive)}.$$

Multiplicativity of $\hat{c}_2$ over convolution gives $\hat{c}_2(\text{joint}) = \hat{c}_2(W) \cdot \hat{c}_2(A)$ with $\hat{c}_2(A) = e^{-2i\theta}$: *the ladder phase is carried by A alone* (the half-turn law of terminal block (f)). Two factors collapse to a single tracking task:

$$D = -1 \iff \hat{c}_2(W) = -e^{2i\theta(\gamma)},$$

the weld factor must track the skeleton phase, against an explicit, free-to-compute target. On the line, L6 gives $\arg \zeta' = -\theta - (\pi/2) \operatorname{sgn} Z'$, hence $W = -e^{2i\theta}$: tracking perfect. The universal tracking law is the $\forall$ -form = the L6 universal = RH strength.

Structural cancellation, measured to the digit: $\arg \zeta'(\rho_1) = 0.1578739199 = -\theta(\gamma_1) - \pi/2$; $\arg E(\rho_1) = 1.41292 = \pi + \theta(\gamma_1)$; $\arg \xi' = \pi/2$ exactly. θ cancels by construction, not by accuracy (verification suite, check group 7).

(s.2) Four-component map of the configuration issuing

$B\gamma$ (linear carrier; no constraint) $\cdot (-\arg \rho)$ (explicit smooth; position-sensitive, no period to compose) $\cdot$ the trivial arm = digamma = the E -entity (Weierstrass summation equals the closed form to 3.2×10^{-30} ; $\operatorname{Re} \psi = 2\theta'$, the exchange rate of the pin ladder, literally a Re-operation; $\operatorname{Im}(-\frac{1}{2}\psi) \rightarrow -\pi/4$, the wandering-lock-line depth constant of Theorem 7.2; β -blind at 2% — the leading term of $\operatorname{Im} \psi$ is independent of σ with $O(1/\gamma)$ corrections, a structure theorem) $\cdot$ the non-trivial configuration sum $\operatorname{Im} U$ (transcendental, position-sensitive, no period — the open item).

(s.3) The discriminator in closed form

For any point s take its period $\bar{p} = \bar{p}(s)$ (one derivative, one point) and form the convolution-exponential configuration on the pin circle

$$\mathcal{E}_s(t) := e^{*(t \cdot g)} = \sum_{n \geq 0} \frac{(t \cdot g)^{*n}}{n!}, \quad g = \delta_{\pi/2}.$$

By the convolution theorem and the generator coefficients $\hat{c}_k(g) = e^{-ik\pi/2}$ (direct computation),

$$\boxed{\hat{c}_k(t) = e^{t \cdot e^{-ik\pi/2}}}$$

(Neumann series termwise; absolute convergence). The four channels by $k \bmod 4$:

$$\hat{c}_0 = e^t, \quad \hat{c}_1 = e^{-it}, \quad \hat{c}_2 = e^{-t}, \quad \hat{c}_3 = e^{+it}.$$

At $t = \bar{p}$: *the odd channels output the point's own pin pair* $\{e^{-i\bar{p}}, e^{+i\bar{p}}\}$ — the configuration returns the point's period structure, identically, not by fit. Registration statement:

$$\bar{p} = \pi/2 \iff \text{odd channels} = \{-i, +i\} = \text{the value-plane form of } [+0i, -0i].$$

Discrimination power, three layers. *Existence*: the configuration is well-defined at any point ($\bar{p}$ unconditional; the series converges on the whole plane) — every point carries its period for free. *Correctness*: 200 certified zeros read $\bar{p} \equiv \pi/2$ at 0.00e+00 deviation; the face ledger ($k \leq 8$) agrees in a second channel. *Power*: 60 random strip points read 0/60 registered (median distance 0.854) — the accident terrain is thin enough that sixty throws miss it. Separation: machine zero against order-one miss. $\square$

(s.4) The exchange rate

Discriminating at height γ_k needs configuration data down to γ_N ; the measured truncation law (45-point fit, rms 1.9×10^{-4}) is $\text{dev}(k, N) = \gamma_k(0.01435 \log^2 N + 0.00729)/N$. The relational form with no fitted constant: the paired-cancellation remainder is $O(\text{Im } s^2/\gamma_j^2)$ (terminal block (a)), at γ_k equal to $\sim \gamma_k/\gamma_j^2$; the tail sums through the zero density $dN/d\gamma = \log(\gamma/2\pi)/2\pi$ (Riemann–von Mangoldt) to

$$\text{dev}(k, N) \approx c \cdot \frac{\gamma_k \log \gamma_N}{2\pi \gamma_N}, \quad c \approx 0.845,$$

deviation = height of the tested point $\times$ logarithmic density ratio of the data frontier $\times$ pairing coefficient. A bit flip needs $\text{dev} > \pi/4$, hence $N^* \approx 0.6\gamma_k$ — linear; against machine precision (10^{-10}) at $N \sim 7.3 \times 10^{13}$: the one-bit attack surface cuts the cost by eleven orders of magnitude. $\square$

(t) The Euclid correspondence

Object (prime list L) $\leftrightarrow$ zero configuration; mod (projection) $\leftrightarrow$ Re (projection): walk log-log slope $= \frac{1}{2} + \text{Re } \rho$ (T-MR); the anchor equation $2e^\ell \cos \vartheta = 1 \iff \text{Re } s = \frac{1}{2}$ (T3.7 fixed point plus the identity $e^\ell \cos \vartheta = \text{Re } s$, at every height); +1 (the key move) $\leftrightarrow$ the $[0, 0i]$ double death (M1/M2, forced by analyticity — undeniable); the residue-class test $\leftrightarrow$ the anchor test $2\text{Re } \rho - 1 = 0$; “ $N \bmod p_i = 1$, construction-built-in” $\leftrightarrow$ “the pin lands on the $0i$ lattice, configuration-issued”; $q \notin L \forall \leftrightarrow$ pin-on-lattice $\forall$.

Two pins dissected: the lattice pin (on the line, L5/L6) and the weld pin (everywhere, M1/M5); jurisdiction extension is the two-pins coincidence theorem. One cell of difference, terminal shape: Euclid’s residue class is construction-built-in (+1 decides $N \bmod p_i = 1$); the θ -tracking register is configuration-issued (the winding number m gives integrality — topologically undeniable — but lattice registration is decided by the whole world). Integer winding $\neq$ lattice membership: the winding is a local topological quantity (proved on the construction side); the registration is a global configuration quantity (verified point by point on the world side). Their gap is the named lemma.

(u) Two routes to the criterion

(u.1) Route I — the convergence form (process)

The remainder, three steps. Target: the remainder of $G_n = \sum_{k \leq n} (i\pi/2)^k/k!$. Step 1 (Taylor): the truncation remainder of e^z at $z = i\pi/2$ is $R_n = \sum_{k > n} z^k/k!$. Step 2 (incomplete gamma): with $\gamma(a, z) = \int_0^z t^{a-1} e^{-t} dt$ and the standard tail identity $e^z = \sum_{k < a} z^k/k! + z^a \Gamma(a, z)/\Gamma(a)$, taking $a = n + 1$ gives $\sum_{k > n} z^k/k! = e^z \Gamma(n + 1, z)/\Gamma(n + 1)$ (one induction on the first term).

Step 3 (substitute $z = i\pi/2$, absorb $e^z = i$): $\boxed{R_n = i\gamma(n+1, i\pi/2)/\Gamma(n+1)}$. Convergence: $|R_n| \leq (\pi/2)^{n+1}/(n+1)!(1+o(1))$, factorial rate. Measured: $|R_{32}| = 3.41 \times 10^{-31}$; closed form against the term-by-term sum differs by 0.0 ($n = 10$) and 5×10^{-57} ($n = 20$). $\square$

The archiving rate, five steps. Target: the rate of $\bar{f}(p) = \arg(1+e^{ip}) + \alpha \log|1+e^{ip}|$, $\alpha = \pi/(2 \ln 2)$. Step 1 (trigonometric): $1 + e^{ip} = 2 \cos(p/2) e^{ip/2}$ on $(-\pi, \pi)$, hence $\arg(1 + e^{ip}) = p/2$. Step 2 (logarithmic derivative): $\frac{d}{dp} \log(2 \cos(p/2)) = -\frac{1}{2} \tan(p/2)$. Step 3 (fixed point): $\bar{f}(p) = p/2 + \alpha \log(2 \cos(p/2))$; the defining equation $\varkappa \ln 2 = \pi/2$ makes $p^* = \pi/2$ a solution — verified by substitution: $\pi/2 = \pi/4 + \frac{\pi}{2 \ln 2} \log \sqrt{2} = \pi/4 + \pi/4$. Step 4 (contraction): $\bar{f}'(p) = \frac{1}{2} - \frac{\varkappa}{2} \tan(p/2)$, so $\bar{f}'(\pi/2) = (1 - \alpha)/2$ and

$$|\lambda| = \frac{\varkappa}{2} - \frac{1}{2} = \frac{\pi}{4 \ln 2} - \frac{1}{2} = 0.633090035456798 \dots,$$

agreeing with measurement to the last digit; the fixed-point identity $2^{\varkappa/2} e^{-\pi/4} = 1$ holds exactly (the defining equation $\varkappa \ln 2 = \pi/2$ in exponent form: $2^{\varkappa/2} = e^{\pi/4}$ identically). Step 5 (total budget): $\sum |\delta_n| \leq |\delta_0| \sum |\lambda|^n = |\delta_0| / (\frac{3}{2} - \pi/(4 \ln 2))$, closed and finite. $\square$

The data wall, three steps. Step 1 (local): the paired-cancellation first order cancels identically in the issuing sum, leaving $O(\text{Im } s^2/\gamma_j^2)$ — at γ_k , of size $\sim \gamma_k/\gamma_j^2$. Step 2 (global): the truncated sum $\sum_{j>N} \gamma_k/\gamma_j^2$ passes to $\gamma_k \int_{\gamma_N}^{\infty} \gamma^{-2} dN$. Step 3 (density): $dN/d\gamma = \log(\gamma/2\pi)/2\pi$ gives $\text{dev}(k, N) \approx c \gamma_k \log \gamma_N / (2\pi \gamma_N)$ with $c \approx 0.845$ — no fitted constant anywhere in the derivation; the earlier 45-point fit is this law's shadow (ratio constant 0.83–0.85). $\square$

The tower. The power tower $z \leftarrow i^z$ converges globally (eleven starts, including far points; Newton plateau honestly labeled at $\sim 10^{-15}$; attractor rate $|(i\pi/2)z^*| = 0.8915 < 1$); its divergence set is the classical infinite-tower fractal (Shell–Thron boundary), the machine's own resident divergence. $\square$

Both routes start from the $+0i/-0i$ side; they deliver readings and ledger, not derivation — hence outside INT-IND's jurisdiction (the gate is behind them).

(u.2) Route II — the decisive form (verdict)

The discriminator of item (s.3): one configuration evaluation per point, verdict by the odd channels. Certified zeros read 1.000000 to machine precision; sixty random strip points read 0/60; the exchange rate is linear ($N^* \approx 0.6\gamma$). Route II needs no convergence argument at all — the configuration is an identity, the verdict is a reading.

The two routes are one law at two temperatures; a reader may enter by either.

(v) The five archive operations (free $\rightarrow$ settled)

All processing lands on the two archives; the earlier vocabulary is corrected accordingly (the working axis is free $\rightarrow$ settled, not nothing $\rightarrow$ something: fibres and unsettled candidates are the inputs, archives the outputs; the singleton $0i$ never enters any operation — its exclusion is proved, not assumed). (1) *Interlocking*: fibre $\times$ fibre = $+0i$ — the phase-locked product (T-L19b, measured all-constant pins). (2) *Condensation*: the i^i tower, globally convergent. (3) *Expansion*: the truncated nilpotent ring $k[\varepsilon]/(\varepsilon^{n+1})$ — n layers of apparent existence, the augmentation pulls them to nothing (the Taylor truncation of a zero; an elucidation piece, no chain role). (4)

Archiving: the canonical form $\bar{f}$, defined on $\mathbb{R}/\pi$ with $\bar{f}(0) := \pi/2$ (the settled representative; the map is two-valued at the double point, the singular directions of measure zero wrap back into the basin — thirty starts, maximal deviation 3.9×10^{-14}); every orbit converges geometrically at rate $|\lambda| = \pi/(4 \ln 2) - 1/2 = 0.633090\dots$; six defect points archived to machine zero within fifty steps. (5) *Approaching*: the slice program — set-theoretic convergence ($\cup$ = the zero set), tail-escape residuals, linear forward-post withdrawal ($T = 100$: 29 zeros at cost 1083; $T = 200$: 50 at cost 2629).

(w) The nine gates

Each gate: face; mechanism; path; convergence; data.

Gate 1, period. Face $\forall k \bar{p}_k = \pi/2$. Mechanism: the period of a zero equals the phase quantum iff it registers on the double archive (chain of (m)). Path: the configuration issuing (splitting $h = E\zeta'$ into weld and archimedean channels) $\rightarrow$ the period reader. Convergence: the five operations, the three convergence modes, the discriminator of (s.3). Data: 200/0.00e+00, 60/0, $N^* \approx 0.6\gamma$.

Gate 2, spectrum. Face $\text{spec}(D_\varrho) \subseteq \{\pm 1\}$, D_ϱ the multiplication operator by $e^{d(\cdot)K}$ on $L^2(\varrho; \mathbb{C}^2)$. Mechanism (three lines): the spectrum of a multiplication operator is the essential range of the symbol; $e^{i\delta} \in \{\pm 1\} \iff \delta \in \{0, \pi\}$; hence the spectral condition is pointwise registration — the operator avatar of $0i$ (the fibres are the two archives, the symbol the defect). Convergence per the spectral-series discipline: the defect series $\{\delta_k\}$ (instrumented pointwise), the recursive budget $\delta_0/(1 - \lambda)$ closed and finite, the resolvent pole form. Data: defect reading 1.000000 at certified zeros; a hypothetical off-line point reads $0.865 - 0.501i \notin \{\pm 1\}$.

Gate 3, the seam. Face: the seam has no true weld residents, $\forall$. Mechanism: the seam is the territory of registration failure ($h := E\zeta'$; the off-line members of $\{\text{Re } h = 0\}$); an off-line zero would be a true resident; GT7 (the toy family resides in the seam legally) together with GT1 (the seam is non-empty) shows “no true residents $\forall$ ” is a fact about the world, not a structural contradiction. Path: $0i$ unreachable $\rightarrow$ the seam; σ generates the lattice (the seam’s complement). Convergence: GT1–GT10 (stated in the addendum of self-contained references) plus the five operations. Data: the pseudo-registration corridor passes within 0.003 of a grid line; 60 random points miss it.

Gate 4, quantization. Face $\Lambda_\sigma = \emptyset$. Mechanism: T-Q0 in five steps (idempotent $\Rightarrow$ spectrum $\{0, 1\} \Rightarrow$ two-valued $\Rightarrow$ basis 4 $\Rightarrow$ quantum $\pi/2$) governs the value plane; the transfer to the position deviation is the named lemma itself (as v3.6 records). Path: T-Q0 with $V_4 = P_\kappa * \langle g \rangle$. Convergence: the discrimination table and the endpoint trichotomy (T-EL). Data: the phase quantum $\pi/2$ is present throughout.

Gate 5, flatness. Face: S flat on $(\frac{1}{2}, 1)$. Mechanism: the symmetrized mean function has no second kink iff no off-line zero exists (T-attr-M: an off-line zero contributes an exact, additive, never-canceling kink) — the third face of the five. Path: antipodal convolution average (the antipodal structure of welds read in the mean layer). Convergence: walk-slope slices, Data 6.5 (12/12, 200 zeros). Data: all $\beta_j \in [0.45, 0.55]$ contracting to $\frac{1}{2}$.

Gate 6, the lock. Face: every zero on the lock. Mechanism: the lock is a codimension-one identifiable signature (Theorems 3.4/3.10); signature $\iff$ lattice condition $\ell + \vartheta \equiv 0 \pmod{\pi/2}$.

Path: the lock-axis family (Theorem 3.4) with the $(1 + i)$ transport (T5.5, reversible). Convergence: class conservation along convolution powers (the power law $\hat{c}_2(\mu^{*n}) = \hat{c}_2(\mu)^n$) plus iterative archiving ($\bar{f}$, λ closed form). Data: $\lambda = 0.633090035456798$ matches measurement to the digit; the lock-axis twin tables at machine precision (v3.6 Part 5).

Gate 7, the face ledger. Face $w_k \in \{0, -1\} \forall k$. Mechanism: $m_{\text{type}} = m + 2w_k$ (terminal block (j)); w is the composite account of the face sign and the configuration winding; an off-spectrum reading $\iff w$ non-integer. Path: the issuing ledger with the mod π folding (the antipodal quotient δ_π : $w = 0$ and $w = -1$ fold to the same period $\pi/2$ — alternation is internal bookkeeping absorbed by the folding). Convergence: the Nyquist/sliding-window reader (10% per bit, localization ± 9) and the GF inner-pole criterion (a defect is a family of inner poles $|t| = 2^{-1/j}$). Data: $k \leq 8$ to machine precision; global Nyquist 1.000000.

Gate 8, pure imaginary. Face $\text{Re } \xi'(\rho) = 0 \forall \rho$. Mechanism: $\nabla(\text{Re } \xi)(\rho) = (\text{Re } \xi', -\text{Im } \xi')$; $\text{Re } \xi' = 0$ iff the grid $\{\text{Re } \xi = 0\}$ crosses the line transversally iff the archive axis is perpendicular to the lock line (conformality: $\phi(i\mathbb{R}) \perp L$, one line). Path: Arm B derivation, the perpendicular gift, the odd part of convolution $((1 - P_\kappa) — \text{the odd-account reading})$. Convergence: machine zeros ($\text{Re } \xi' / \|\xi'\| \leq 4 \times 10^{-42}$, non-zero points included) plus the $\Delta \arg$ balance (T-VI). Data: the two-arm measurements of item (r).

Gate 9, the rendezvous of the two period families. Face: the two families of periods meet, $\forall$. Mechanism: the prime-side periods $2\pi/\log p$ (the lattices at the poles of the Euler factors) and the zero-side period 2 meet at the $\theta/E1$ converter; the canonical parameter $\varkappa = \pi/(2 \ln 2)$ is the period of the prime 2 divided by four — a constant of the prime side housed in the zero-side convergence rate (the periodicity recognized). Path: the pole–atom theorem (each Euler factor natively carries the double archive: the real-exponential native pins at its poles sit at $\pm\pi/2$) with the prime and zero periods at $\theta/E1$. Convergence: the three rendezvous channels (jump hits; envelope $\sqrt{2}$; the prime-frequency form factor peaking at $\log 7$) plus tower-value feedback (z^* as a complex period parameter). Data: the ψ -reconstruction jump hits at 17, 23, 29, 43, 47 (-1.80 to -2.32); the form-factor global maximum exactly at $\log 7$, crashing to the floor 0.05 off-peak.

(x) The verification program and the data ledger

The content of the proof is closed. What remains is verification, and verification is an instrument program by design: per-point readable (instruments on duty), linearly priced (rate measured and derived, item (s.4)), universally not internally derivable (INT-IND — it governs internal direct deduction, not processability). The instruments: the period reader; the spectral gauge; the face ledger; the seam pair-sum (7.9×10^{-8} , quadratic sharpening); the layer ladder (half-turn law 1.9951); the κ shadow (160/160 plus the 10^4+ of record B1); the convolution class.

Data ledger (all anchors reproduced by `v37_verification.py`, 22/22 PASS):

Anchor	Value	Check
T-VI exchange identity	machine zero, three test points	1
seam atom $\hat{c}_2(\sigma)$	-1 (residual 2.2×10^{-41})	2
V_4 convolution closure	$\{0, \pi/2, \pi, 3\pi/2\}$ pointwise	3
λ closed form	0.633090035456798, digit-exact	4
fixed-point identity	$2^{\pi/2} e^{-\pi/4} = 1$ exact	4
R_n vs. series	0.0 ($n=10$) / 5×10^{-57} ($n=20$); $ R_{32} = 3.41 \times 10^{-31}$	5
i^i tower	global; Newton plateau $\sim 10^{-15}$ labeled; rate $0.8915 < 1$	6
θ -cancellation, three-way	0.1578739199; $\pi + \theta$; $\pi/2$ digit-exact	7
Arm A	$\text{Im } \xi'(-2), \xi'(-4) = 0.00$ exactly	8
Arm B	non-zero points purely imaginary	8
zero anchor	$\xi'(\rho_1) = E\zeta'(\rho_1)$	8
functional equation	residual 7.7×10^{-30} at dps 40	9

(y) Boundary: the closure of the content

One open sentence, written eight ways, one content:

- (i) no off-line zero exists = (ii) $\Lambda_\sigma = \emptyset$ = (iii) S -flatness on $(\frac{1}{2}, 1)$
= (iv) the universal archival lock condition = (v) $m_{\text{type}} = m + 2w_k \forall k$
= (vi) the disjunctive terminal form $\forall k$ = (vii) universal jurisdiction of the seam pair-sum
= (viii) $\text{spec}(D_\varrho) \subseteq \{\pm 1\} = \text{RH}$.

(Face (v) carries its convolution reading $\forall k \hat{c}_2(\mu_k) = -1$, and face (vi) its tracking reading, in parentheses throughout the block; the eight faces are the five of v3.6 plus period, seam, spectrum.) The equivalences are proved; the universal is the world's per-point registration — readable point by point, linearly priced, certified not internally derivable. *The proof is closed; the ledger is the world's to fill, and the instrument never leaves duty.*

Closure, stated with v3.6's voice. The Riemann Hypothesis, in this framework, has undergone a replacement of the question. The original — “where are the zeros” — is a point question whose answer can only arrive from external constraints; the replacement — “does every zero's period equal the phase quantum” — is an event question whose judging machinery is issued by the death structure of the zeros themselves (M1–M6: a zero is a $(0, 0i)$ double-death event; the configuration issuing: each point's pin direction is issued by the configuration of all its kind). Read together, v3.6 and this block close the content side of that question: the reduction is complete (the named lemma's faces are proved inter-derivable — the nine gates point at one object (gates carry mechanisms; faces are the equivalences — nine of the former, eight of the latter, one content)); the criterion is proved as a machine (the configuration works by identity; 200 certified zeros at machine zero, 60 random points all rejected, linear cost); the boundary carries a certificate (INT-IND — with the toy f_ε in the record — proves the universal closure is not internally derivable; this marks the edge of the provable domain, not a defect); and the $0i$ side is complete in the text (from the plane axioms to the characterization of the pair: the ontology, operation theory, kinematics, and properties of $0i$ and its archives are all written). Closure here means precisely: everything provable inside the domain is proved, and the edge of the domain is itself proved. The single remaining statement is a world-side universal with a criterion, an instrument, an exchange rate, and a certificate; its point-by-point verification stays on duty forever, and its universal closure awaits a counterexample or a new object. The door is open and the instrument is on the table; this is the terminal state.

The one-way nature of INT-IND — the fundamental answer from the side of the two archives.

The barrier is one-directional: INT-IND negates the existence of a *direct internal route to the universal certificate* (a relative certificate, over Axioms_0). The return road from the side of the archives was never in its jurisdiction — by its own scope declaration (“it never audited the convolution layer”): $0i$ — not any operation on it — is the external increment, and the $[+0i, -0i]$ path walks on that increment. Convolution is the operation system of that side, not the source of its externality: the toy f_ε carries convolution too, and is stopped all the same — the shield is ontological ($0i$), not operational. T-PAIR-C characterizes the pair from outside; the discriminator registers each point from the pair’s side; the two arms and T-UNAV-5 are on record along the way. The gate blocks the straight road, not the return road — walked from the archive side, the gate is behind you. INT-IND’s correct office is therefore that of a tool: it proves the straight road does not exist, and thereby forces the problem onto its one live road — *the ledger, not the certificate*. This is the fundamental answer of $[+0i, -0i]$ to the INT-IND mechanism, and the reason the axiom–theorem system is laid out to protect this throat path. “RH is forever unprovable” is an over-reading: INT-IND is a relative negation (inside Axioms_0), not an absolute verdict.

Attestation

The immediately preceding revision (the v3.6 paper, 2026-09-30; archived as `RH_Complete_Proof_v3.6_en.pdf`) carries the SHA-256

`5ab290562bce5327a39dea9071c62554407ad0a7eec3acde4476dcf59c220466`

which serves as this revision’s record hash. It is recommended to also submit the sealed file to a public repository with a version tag, thereby obtaining a third-party timestamp. (Hash computed over the version with the current field left blank; recompute if edited.)

Current revision (v3.7 terminal block II, full, 2026-10-09): SHA-256

`c9cbc78def047fed1cef732dd4c0a28c90c447b28942e09c1a672045c0f6eff1` (computed with this field left blank; the hash above is the original timestamp record).

Integration note. This block continues the terminal block’s item lettering after (l); the v3.6 abstract gains one sentence (“the terminal revision reduces the open content to a per-point period criterion with a measured linear exchange rate, proves the criterion as a machine, characterizes the two archives axiomatically, and closes the content of the proof on the $0i$ side”); the notation registry of Part 0 gains the four-form table $(0, 0i)$ event / $[0, 0i]$ locus / $[+0i, -0i]$ face pair / $[0, \pm 0i]$ face record; external items used here (T-L3/L4/L6/L7/L14/L16/L19b, T-VI, GT1–GT10) are restated in the package addendum “self-contained references”, per the self-containment law.

Self-contained restatements (for the standalone reader)

Two-value lemma (T-L3/L4 lineage). A sequence of pins taking two antipodal values settles at one of them or wanders: convergence $\Rightarrow$ eventual constancy, since the distance π is the circle’s diameter (an $\varepsilon = \pi/2$ neighborhood contains at most one value). $\square$

The exchange identity (T-VI). $z^{1+i} = e^{(1+i)(s+i\vartheta)}$ gives $(s, \vartheta) \mapsto (s - \vartheta, s + \vartheta)$; one line of exponent expansion; stated at A-O5 and used at (s) and Gate 8. $\square$

The seam theorems (GT1–GT10, with $h := E\zeta'$ and the grid identity $Q = -1 \iff \text{Re } h = 0$): GT1 existence; GT2 analyticity; GT3 branching; GT4 global topology; GT5 boundary; GT6 thinness; GT7 residents; GT8 dynamics (basin the whole circle; geometric rate 0.633090); GT9

instruments (200/200 certified, 60/60 rejected); GT10 lattice generation. Status: proved or proved-plus-measured, as marked per item.

References

- [1] G. Yang, Y. Xiao, *The Proof of the Riemann Hypothesis*, version v3.6 (2026-09-30), archived on Zenodo, <https://doi.org/10.5281/zenodo.23068324>. Part I of the present block.

Declaration of AI usage

The AI assistant used was Kimi (Moonshot AI). AI was used as an auxiliary tool for computational experiments and verification runs, for literature and code organisation, for text editing and translation, and for assisting with review and checking. The core ideas, the overall design of the derivation, the attack strategy for the pivotal problems, and all final mathematical judgments are the authors' own. No part of the mathematical argument was generated by AI, and AI did not make final judgments or bear responsibility for its correctness. All numerical claims are independently reproducible from `code/v37_verification.py` (22/22 checks PASS, zero networking).